

BIODIVERSITY LEVELS AND HOTSPOTS

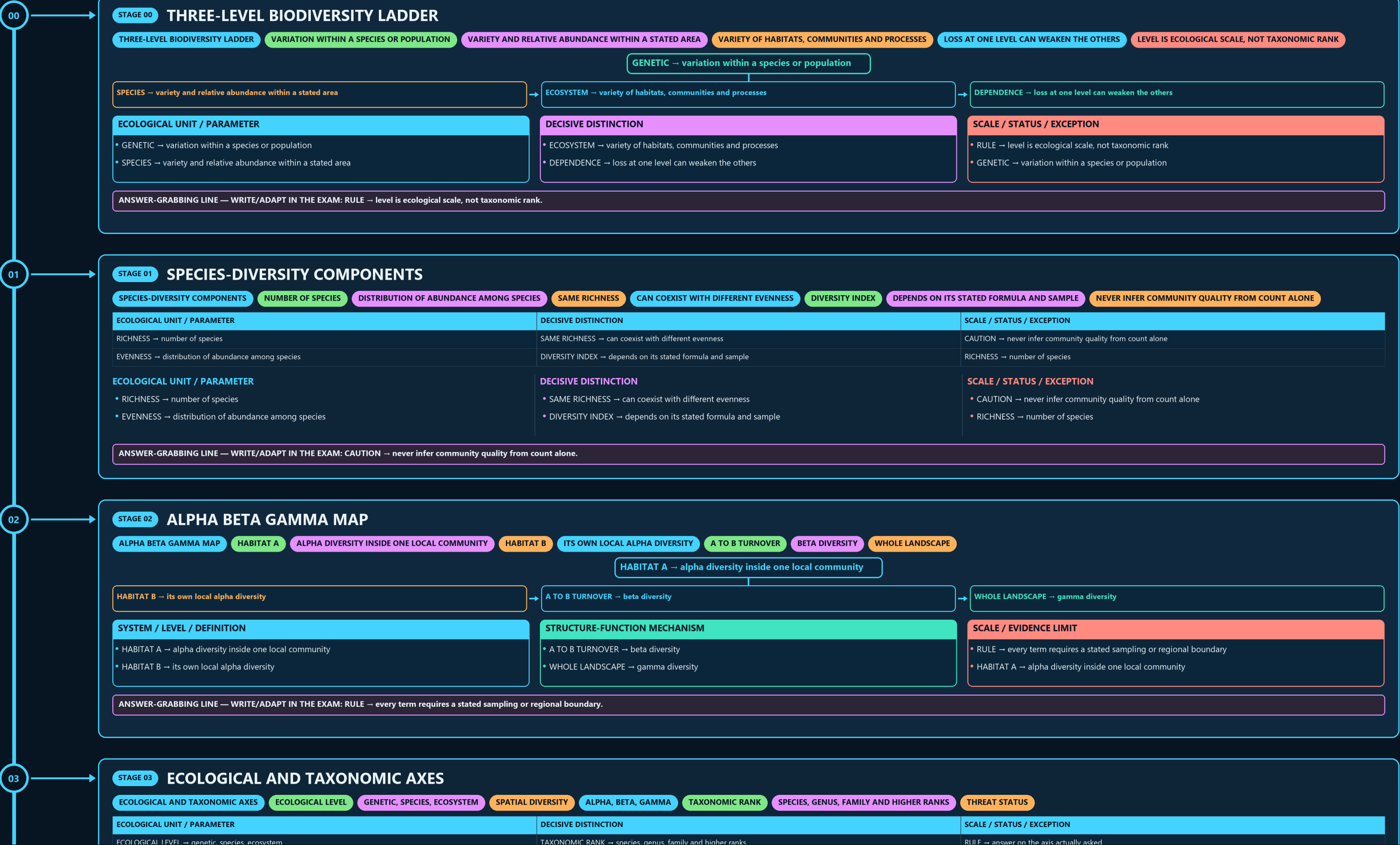
THREE-LEVEL BIODIVERSITY LADDER → SPECIES-DIVERSITY COMPONENTS → ALPHA BETA GAMMA MAP → ECOLOGICAL AND TAXONOMIC AXES → DISTRIBUTION VOCABULARY → HOTSPOT CRITERIA GATE

Read the thick cyan rail from Stage 00 to the final core synthesis. Each card uses a concept-appropriate structure; the grey E card is optional enrichment only and never repairs missing core content.

PRIMARY CORE FLOW SUBORDINATE EXTRA APPROVAL / REVIEW

Approval: FALSE • Pending user review • active generation g1 • explicit approval of this exact generation is required • all prior artifacts unchanged

CORE BEFORE EXTRA • prior artefacts unchanged • exact same master drives poster and tiles



03

SYSTEM / LEVEL / DEFINITION

- HABITAT A → alpha diversity inside one local community
- HABITAT B → its own local alpha diversity

STRUCTURE-FUNCTION MECHANISM

- A TO B TURNOVER → beta diversity
- WHOLE LANDSCAPE → gamma diversity

SCALE / EVIDENCE LIMIT

- RULE → every term requires a stated sampling or regional boundary
- HABITAT A → alpha diversity inside one local community

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → every term requires a stated sampling or regional boundary.

03

STAGE 03

ECOLOGICAL AND TAXONOMIC AXES

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ECOLOGICAL LEVEL

GENETIC, SPECIES, ECOSYSTEM

SPATIAL DIVERSITY

ALPHA, BETA, GAMMA

TAXONOMIC RANK

SPECIES, GENUS, FAMILY AND HIGHER RANKS

THREAT STATUS

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
ECOLOGICAL LEVEL → genetic, species, ecosystem	TAXONOMIC RANK → species, genus, family and higher ranks	RULE → answer on the axis actually asked
SPATIAL DIVERSITY → alpha, beta, gamma	THREAT STATUS → separate assessment of extinction risk	ECOLOGICAL LEVEL → genetic, species, ecosystem

ECOLOGICAL UNIT / PARAMETER

- ECOLOGICAL LEVEL → genetic, species, ecosystem
- SPATIAL DIVERSITY → alpha, beta, gamma

DECISIVE DISTINCTION

- TAXONOMIC RANK → species, genus, family and higher ranks
- THREAT STATUS → separate assessment of extinction risk

SCALE / STATUS / EXCEPTION

- RULE → answer on the axis actually asked
- ECOLOGICAL LEVEL → genetic, species, ecosystem

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE → answer on the axis actually asked.

04

STAGE 04

DISTRIBUTION VOCABULARY

DISTRIBUTION VOCABULARY

OCCURS NATURALLY IN THE DEFINED REGION

NATURALLY RESTRICTED TO THE DEFINED REGION

LOW ABUNDANCE OR OCCURRENCE UNDER A STATED MEASURE

STATUS UNDER AN ASSESSMENT FRAMEWORK

ONE SPECIES MAY SATISFY SEVERAL, BUT TERMS ARE NOT

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
NATIVE → occurs naturally in the defined region	RARE → low abundance or occurrence under a stated measure	CAUTION → one species may satisfy several, but terms are not synonyms
ENDEMIC → naturally restricted to the defined region	THREATENED → status under an assessment framework	NATIVE → occurs naturally in the defined region

ECOLOGICAL UNIT / PARAMETER

- NATIVE → occurs naturally in the defined region
- ENDEMIC → naturally restricted to the defined region

DECISIVE DISTINCTION

- RARE → low abundance or occurrence under a stated measure
- THREATENED → status under an assessment framework

SCALE / STATUS / EXCEPTION

- CAUTION → one species may satisfy several, but terms are not synonyms
- NATIVE → occurs naturally in the defined region

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CAUTION → one species may satisfy several, but terms are not synonyms.

05

STAGE 05

HOTSPOT CRITERIA GATE

HOTSPOT CRITERIA GATE

GATE 1

AT LEAST 1,500 ENDEMIC VASCULAR PLANT SPECIES

GATE 2

30 PER CENT OR LESS OF ORIGINAL NATURAL VEGETATION

BOTH GATES MET

HOTSPOT PRIORITISATION FRAMEWORK APPLIES

FAILED GATE

GATE 1 → at least 1,500 endemic vascular plant species

→

GATE 2 → 30 per cent or less of original natural vegetation remains

→

BOTH GATES MET → hotspot prioritisation framework applies

→

FAILED GATE → species richness alone cannot create hotspot status

SYSTEM INPUT / STARTING CONDITION	ECOLOGICAL PROCESS / TRANSFER	OUTCOME / FEEDBACK / LIMIT
- GATE 1 → at least 1,500 endemic vascular plant species - GATE 2 → 30 per cent or less of original natural vegetation remains	- BOTH GATES MET → hotspot prioritisation framework applies - FAILED GATE → species richness alone cannot create hotspot status	- GATE 1 → at least 1,500 endemic vascular plant species - GATE 2 → 30 per cent or less of original natural vegetation remains

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FAILED GATE → species richness alone cannot create hotspot status.

06

STAGE 06

DESIGNATION FIREWALL

DESIGNATION FIREWALL

CONSERVATION INTERNATIONAL SCIENTIFIC PRIORITISATION FRAMEWORK

PROTECTED AREA

LEGAL DESIGNATION UNDER APPLICABLE DOMESTIC LAW

BIODIVERSITY HERITAGE SITE

SEPARATE STATUTORY CATEGORY

ECO-SENSITIVE ZONE

NOTIFICATION-BASED REGULATORY STATUS

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
HOTSPOT → Conservation International scientific prioritisation framework	BIODIVERSITY HERITAGE SITE → separate statutory category	RULE → one label never automatically confers another status
PROTECTED AREA → legal designation under applicable domestic law	ECO-SENSITIVE ZONE → notification-based regulatory status	HOTSPOT → Conservation International scientific prioritisation framework

ECOLOGICAL UNIT / PARAMETER

DECISIVE DISTINCTION

SCALE / STATUS / EXCEPTION

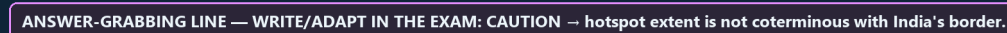
BIODIVERSITY LEVELS AND HOTSPOTS • TILE 2/4 • same master rows 2474-5708 of 10656 • continues below

- ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: FAILED GATE → species richness alone cannot create hotspot status.**

NOTIFICATION-BASED REGULATORY STATUS

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: RULE — one label never automatically confers another status.

HOTSPOT EXTENT IS NOT COTERMINOUS WITH INDIA'S BORDER



USE HOTSPOTS FOR PRIORITY, NOT FOR EXCLUSION **NATIONAL STRATEGY MUST RETAIN A WIDER ECOLOGICAL BASELINE**

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → national strategy must retain a wider ecological baseline.

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
• MEGADIVERSE COUNTRY → national biodiversity descriptor • LMMC → international political grouping	• HOTSPOT → transboundary scientific priority region • NBA, SBB, BMC → domestic statutory governance levels	• RULE → four categories answer different questions • MEGADIVERSE COUNTRY → national biodiversity descriptor

09

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: VERDICT → national strategy must retain a wider ecological baseline.

STAGE 09

COUNTRY, LABEL AND INSTITUTION

COUNTRY, LABEL AND INSTITUTION

MEGADIVERSE COUNTRY

NATIONAL BIODIVERSITY DESCRIPTOR

INTERNATIONAL POLITICAL GROUPING

TRANSBOUNDARY SCIENTIFIC PRIORITY REGION

NBA, SBB, BMC

DOMESTIC STATUTORY GOVERNANCE LEVELS

FOUR CATEGORIES ANSWER DIFFERENT QUESTIONS

ECOLOGICAL UNIT / PARAMETER	DECISIVE DISTINCTION	SCALE / STATUS / EXCEPTION
MEGADIVERSE COUNTRY → national biodiversity descriptor	HOTSPOT → transboundary scientific priority region	RULE → four categories answer different questions
LMMC → international political grouping	NBA, SBB, BMC → domestic statutory governance levels	MEGADIVERSE COUNTRY → national biodiversity descriptor

ECOLOGICAL UNIT / PARAMETER

• MEGADIVERSE COUNTRY → national biodiversity descriptor

• LMMC → international political grouping

DECISIVE DISTINCTION

• HOTSPOT → transboundary scientific priority region

• NBA, SBB, BMC → domestic statutory governance levels

SCALE / STATUS / EXCEPTION

• RULE → four categories answer different questions

• MEGADIVERSE COUNTRY → national biodiversity descriptor

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: NBA, SBB, BMC → domestic statutory governance levels.

10

STAGE 10

ABS GOVERNANCE RAIL

ABS GOVERNANCE RAIL

BIOLOGICAL RESOURCE OR ASSOCIATED KNOWLEDGE

ACCESS REQUEST

RELEVANT AUTHORITY

SCRUTINY UNDER THE LEGAL ARCHITECTURE

COMMERCIAL OR RESEARCH USE

APPLICABLE CONDITIONS

BENEFIT SHARING

BIOLOGICAL RESOURCE OR ASSOCIATED KNOWLEDGE → access request

→

RELEVANT AUTHORITY → scrutiny under the legal architecture

→

COMMERCIAL OR RESEARCH USE → applicable conditions

→

BENEFIT SHARING → intended return to eligible claimants or communities

→

PBR → local documentation; not proof that benefits were delivered

SYSTEM INPUT / STARTING CONDITION

• BIOLOGICAL RESOURCE OR ASSOCIATED KNOWLEDGE → access request

• RELEVANT AUTHORITY → scrutiny under the legal architecture

ECOLOGICAL PROCESS / TRANSFER

• COMMERCIAL OR RESEARCH USE → applicable conditions

• BENEFIT SHARING → intended return to eligible claimants or communities

OUTCOME / FEEDBACK / LIMIT

• PBR → local documentation; not proof that benefits were delivered

• BIOLOGICAL RESOURCE OR ASSOCIATED KNOWLEDGE → access request

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: PBR → local documentation; not proof that benefits were delivered.

11

STAGE 11

GLOBAL AND PYQ BOUNDARY

GLOBAL AND PYQ BOUNDARY

KMGBF TARGET 3

AREA-BASED CONSERVATION AND OECMS IN THE OWNER

CBD GUIDANCE

DOES NOT REPLACE OR QUALIFY ADOPTED COP DECISIONS

MASS EXTINCTION ROUTE

HUMAN DRIVERS, NO INFERRED OBJECTIVE ANSWER

ISSG AND EU NRL

KMGBF TARGET 3 → area-based conservation and OECMs in the owner

→

CBD GUIDANCE → does not replace or qualify adopted COP decisions

→

MASS EXTINCTION ROUTE → human drivers, no inferred objective answer

→

ISSG AND EU NRL → external institutional or legal comparators

→

CLOSE → scientific label, domestic law and external comparator stay distinct

ECOLOGICAL UNIT / PARAMETER

• KMGBF TARGET 3 → area-based conservation and OECMs in the owner

• CBD GUIDANCE → does not replace or qualify adopted COP decisions

DECISIVE DISTINCTION

• MASS EXTINCTION ROUTE → human drivers, no inferred objective answer

• ISSG AND EU NRL → external institutional or legal comparators

SCALE / STATUS / EXCEPTION

• CLOSE → scientific label, domestic law and external comparator stay distinct

• KMGBF TARGET 3 → area-based conservation and OECMs in the owner

ANSWER-GRABBING LINE — WRITE/ADAPT IN THE EXAM: CLOSE → scientific label, domestic law and external comparator stay distinct.

E

EXTRA

SUBORDINATE ENRICHMENT — ONLY AFTER THE COMPLETE CORE

OPTIONAL ADVANCED ENRICHMENT

THE CBD'S KUNMING-MONTREAL GLOBAL BIODIVERSITY FRAMEWORK (ADOPTED AT CBD)

INDIA'S IMPLEMENTATION OF 30X30-TYPE COMMITMENTS INTERACTS DIRECTLY WITH THE

NATIONAL SPECIES-COUNT FIGURES FOR INDIA ARE COMPILED BY BSI/ZSI

ENDEMISM IS SOMETIMES CONFLATED IN POPULAR SOURCES WITH SIMPLE

"MEGADIVERSE COUNTRY" LISTS (ORIGINATING FROM CONSERVATION INTERNATIONAL/UNEP-WCMC

THE BIOLOGICAL DIVERSITY ACT, 2002 ESTABLISHED A THREE-TIER STRUCTURE

NBA (NATIONAL)

OPTIONAL ECOLOGICAL PROCESS DEPTH

• The CBD's Kunming-Montreal Global Biodiversity Framework (adopted at CBD COP15, 2022)

• CAUTION: India's implementation of 30x30-type commitments interacts directly with the protected

• CAUTION: National species-count figures for India are compiled by BSI/ZSI across decades of

• CAUTION: Endemism is sometimes conflated in popular sources with simple rarity; an advanced

SCALE, PARAMETER OR STATUS LIMIT

• CAUTION: "Megadiverse country" lists (originating from Conservation International/UNEP-WCMC

• The Biological Diversity Act, 2002 established a three-tier structure: NBA (national),

• Access and Benefit Sharing (ABS) requires NBA/SBB approval for accessing India's

• The CBD's Kunming-Montreal Global Biodiversity Framework (2022) set the 30x30 target —

COMPARATIVE RESTORATION NUANCE

• KMGBF's 30x30 target explicitly counts "other effective area-based conservation

• CBD COP16 (Cali 2024, resumed Rome February 2025) created the Cali Fund for sharing

• India's four biodiversity hotspots are Himalaya, Indo-Burma, Western Ghats-Sri Lanka and

• NOT: NBA directly regulates access by all Indian citizens/entities. - NBA primarily regulates

OPTIONAL STATUS — This optional depth is unnecessary for a competent core answer; use it only for added nuance after the complete core has been written.

END OF MASTER CANVAS • poster embeds this exact master • tiled pages are overlapping crops of the same pixels

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